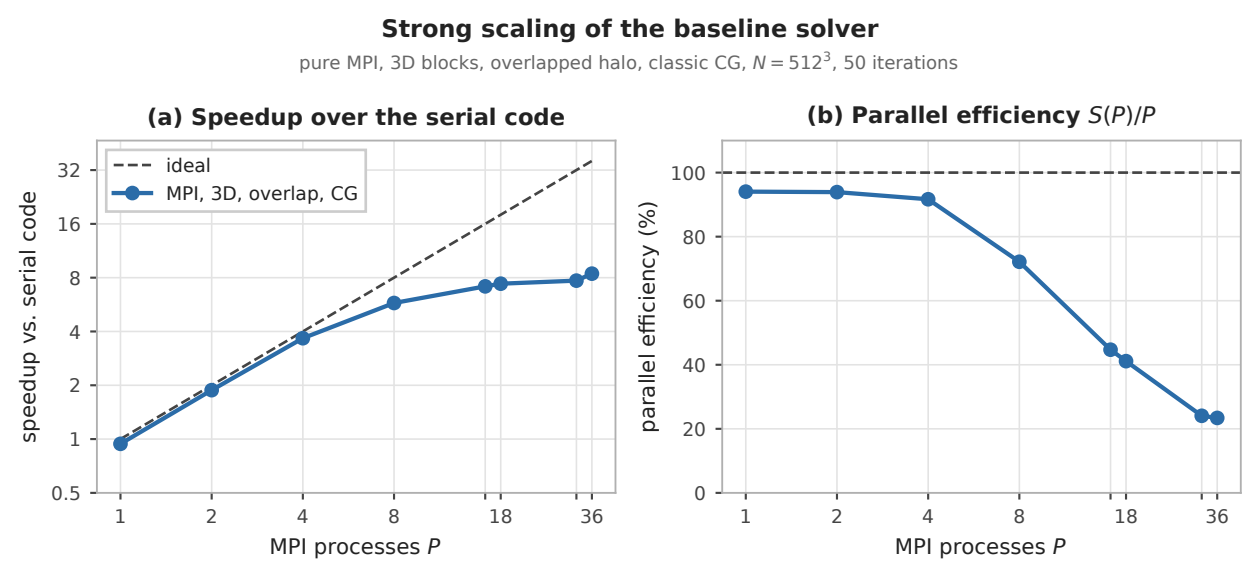
Strong scaling of the baseline solver
pure MPI, 3D blocks, overlapped halo, classic CG, $N = 512^3$, 50 iterations
(a) Speedup over the serial code
ideal
MPI, 3D, overlap, CG
speedup vs. serial code
32
16
8
4
2
1
0.5
1
2
4
8
18
36
MPI processes P
(b) Parallel efficiency $S(P)/P$
parallel efficiency (%)
100
80
60
40
20
0
1
2
4
8
18
36
MPI processes P